Recap of Gaussian processes

Gaussian conditionals

If (X_1, X_2) are jointly Gaussian with distribution

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} A & C \\ C^T & B \end{pmatrix}\right)$$

then the conditional distributions are also Gaussian and given by

$$\begin{aligned} X_1 \,|\, x_2 &\sim N\left(\mu_1 + CB^{-1}(x_2 - \mu_2),\ A - CB^{-1}C^T\right) \\ X_2 \,|\, x_1 &\sim N\left(\mu_2 + C^T A^{-1}(x_1 - \mu_1),\ B - C^T A^{-1} C\right) \end{aligned}$$

The matrix $A - CB^{-1}C^T$ is called the *Schur complement* of B.

Gaussian conditionals

If (X_1, X_2) are jointly Gaussian with distribution

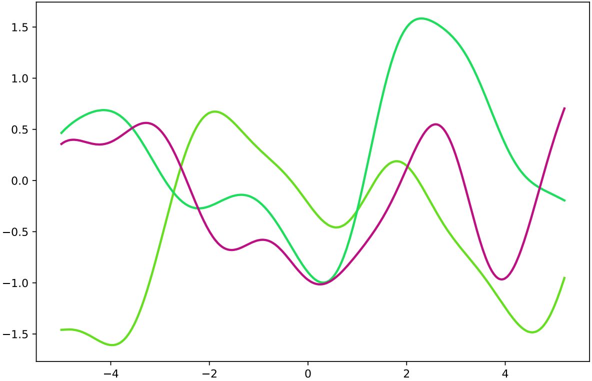
$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix}\right)$$

then the conditional distributions are also Gaussian and given by

$$X_1 \mid x_2 \sim N\left(\frac{K_{12}}{K_{22}} x_2,\ K_{11} - \frac{K_{12}^2}{K_{22}}\right)$$

$$X_2 \mid x_1 \sim N\left(\frac{K_{12}}{K_{11}} x_1,\ K_{22} - \frac{K_{12}^2}{K_{11}}\right)$$

Samples from prior and posterior



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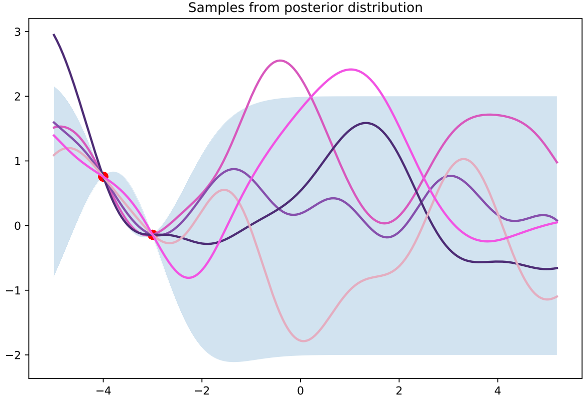
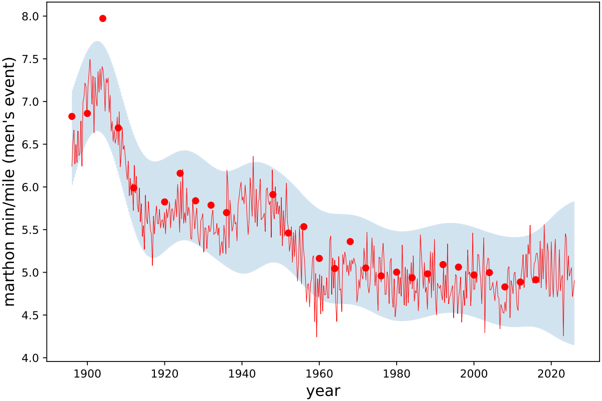


Olympic marathon times (men's race)



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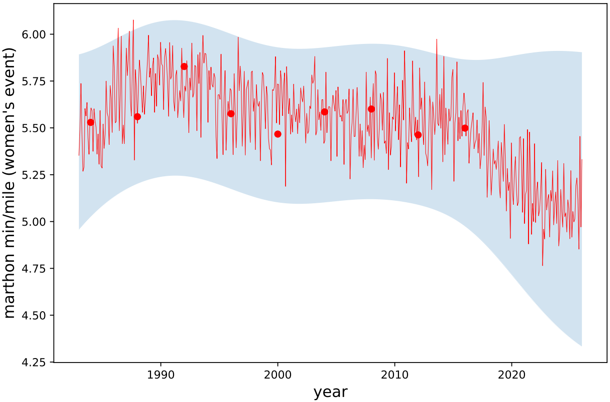

We also briefly introduced the Dirichlet process.

We won’t ask you about this on an exam, but there could be a quiz question on the definition of the Dirichlet process.

DPs vs GPs

The natural correspondence between the Gaussian process and Dirichlet process is as follows:

Gaussian process	Dirichlet process
points $x_1, \ldots, x_n$	sets $A_1, \ldots, A_n$
prior mean μ	prior mean F_0
prior covariance K	prior scaling α

The Dirichlet Process

- The Dirichlet process is analogous to the Gaussian process
- Every partition of sample space has a Dirichlet distribution (more precise shortly)
- GPs are tools for regression functions; DPs are tools for distributions and densities

What exactly is a Dirichlet Process?

Recall:

A random function m is distributed according to a Gaussian process if for every $x_1, x_2, \ldots, x_n$ the random vector $m(x_1), \ldots, m(x_n)$ has a multivariate Gaussian distribution

$$N(\mu(x), K(x))$$

What exactly is a Dirichlet Process?

A random distribution F is distributed according to a Dirichlet process $DP(\alpha, F_0)$ if for every partition $A_1, \dots, A_n$ of the sample space the random vector $F(A_1), \dots, F(A_n)$ has a Dirichlet distribution

$$\mathrm{Dir}\left(\alpha F_0(A_1), \alpha F_0(A_2), \dots, \alpha F_0(A_n)\right)$$

where

$$F(A_i) = \mathbb{P}_F(A_i) = \int_{A_i} dF(x)$$

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The distribution F_0 and scaling α are hyperparameters of the prior

Analogous to the mean μ and covariance kernel K of the GP prior

Examples

As a special case, if the sample space is the real line we can take the partition to be

$$A_1 = \{z \,:\, z \leq x\}$$

$$A_2 = \{z \,:\, z > x\}$$

then

$$F(x) = \mathbb{P}_F(X \leq x) \sim \text{Beta}\Big(\alpha F_0(x), \alpha(1 - F_0(x))\Big)$$

Examples

As another special case, if the sample space is the real line we can take the partition to be

$$A_1 = (-\infty, -5], \;\; A_2 = (-5, 5], \;\; A_3 = (5, \infty)$$

then $(F(A_1), F(A_2), F(A_3))$ is a random point on the 3-simplex

$$\Delta_3 = \{(p_1, p_2, p_3) : p_i \geq 0, p_1 + p_2 + p_3 = 1\}$$

with distribution

$$(F(A_1), F(A_2), F(A_3)) \sim \text{Dirichlet}\big(\alpha_1, \alpha_2, \alpha_3\big)$$

with $\alpha_1 = \alpha F_0(-5)$, $\alpha_2 = \alpha(F_0(5) - F_0(-5))$, and $\alpha_3 = \alpha(1 - F_0(5))$.

Problem from last year's midterm

Suppose that $m(x)$ denotes a function with 1-dimensional input $x \in \mathbb{R}$. We place a Gaussian process prior $\pi(m)$ on such functions with mean function $\mu(x)$ and covariance kernel $K(x, x') = K_\sigma(x - x')$ where K_σ denotes a Gaussian kernel with mean 0 and variance σ^2.

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a) What is the prior mean $\mathbb{E}_\pi(m(x))$ of m at a fixed point $x \in \mathbb{R}$?

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a) What is the prior mean $\mathbb{E}_\pi(m(x))$ of m at a fixed point $x \in \mathbb{R}$?

By the definition of Gaussian process, $m(x) \sim N(\mu(x), K(x, x))$. So, $\mathbb{E}_\pi(m(x)) = \mu(x) = 0$

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b) What is the prior variance $\text{Var}_\pi(m(x))$ of m at a fixed point $x \in \mathbb{R}$?

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b) What is the prior variance $\text{Var}_\pi(m(x))$ of m at a fixed point $x \in \mathbb{R}$?

$Var_\pi(m(x)) = K_\sigma(0) = \frac{1}{\sqrt{2\pi\sigma^2}}$

Problem from last year's midterm

Suppose that $F(x)$ denotes a 1-dimensional distribution function $x \in \mathbb{R}$, so $F(x) = \mathbb{P}_F(X \leq x)$. We place a Dirichlet process prior $\pi(F)$ on such distribution functions with parameters $\alpha > 0$ and F_0 where F_0 is a fixed distribution.

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a) What is the prior mean $\mathbb{E}_\pi(F(x))$ of F at a fixed point $x \in \mathbb{R}$?

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a) What is the prior mean $\mathbb{E}_\pi(F(x))$ of F at a fixed point $x \in \mathbb{R}$?

By the definition of the Dirichlet process,

$$F(x) \sim Beta(\alpha F_0(x), \alpha(1 - F_0(x))).$$

Using the expectation of the Beta, we get $\mathbb{E}_\pi(F(x)) = F_0(x)$.

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b) What is the prior variance $\text{Var}_\pi(F(x))$ of F at a fixed point $x \in \mathbb{R}$?

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b) What is the prior variance $\mathrm{Var}_\pi(F(x))$ of F at a fixed point $x \in \mathbb{R}$?

$$\textit{given: } Var(Beta(\alpha, \beta)) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$Var_\pi(F(x)) = \frac{F_0(x)(1 - F_0(x))}{(1+\alpha)}$$

Summary

- In a Bayesian approach, the parameters are random, and the data are fixed.
- In nonparametric Bayes, the "parameters" are functions
- A Gaussian process is a stochastic process m where each collection of random variables $m(x_1), m(x_2), \ldots, m(x_n)$ is jointly Gaussian
- Calculation with GPs uses Gaussian conditioning via Schur complements
- Gaussian processes are Bayesian versions of kernel regression; the posterior mean is equivalent to Mercer kernel regression